

Adaptive Exam Progression System Architecture

Overview

The CNAHS Examination System is evolving beyond a traditional online quiz platform into an adaptive competency assessment and academic integrity monitoring system.

The purpose of this system is to:

- Measure student competency progressively
- Restrict advancement until mastery is achieved
- Identify weak academic areas automatically
- Track question performance trends
- Provide institutional monitoring and analytics
- Support academic integrity workflows

This architecture separates:

- Faculty academic ownership
 - Student progression and mastery
 - Administrative monitoring and oversight
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Core Philosophy

The system is designed around:

Mastery-Based Progression

Students cannot simply continue to higher difficulty levels automatically.

They must:

- Reach the required passing score
- Demonstrate competency
- Unlock the next difficulty level progressively

This creates:

- Elimination-style progression
 - Adaptive learning reinforcement
 - Stronger academic readiness
 - Better board exam preparation
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Difficulty Progression Structure

Difficulty Levels

EASY
→ MEDIUM
→ HARD
→ EXPERT

Required Passing Rates

Difficulty	Required Passing Rate	Unlocks
EASY	60%	MEDIUM
MEDIUM	70%	HARD
HARD	70%	EXPERT
EXPERT	Completion Tier	Final Mastery

Progression Rules

EASY → MEDIUM

Student must:

- Finish EASY exam
- Achieve at least 60%

If passed:

- MEDIUM unlocks

If failed:

- MEDIUM remains locked
- Student must retry EASY
- Weak topics are recorded

MEDIUM → HARD

Student must:

- Finish MEDIUM exam
- Achieve at least 70%

If passed:

- HARD unlocks

If failed:

- HARD remains locked
 - Student reviews weak areas
-

HARD → EXPERT

Student must:

- Finish HARD exam
- Achieve at least 70%

If passed:

- EXPERT unlocks
-

Core Academic Concepts

Competency-Based Learning

The system evaluates:

- Mastery
- Retention
- Consistency
- Weak areas
- Readiness

instead of only:

- raw exam scores
-

Question Performance Tracking

The system records student performance per question.

This is one of the most important parts of the architecture.

Mastery Categories

LOW Mastery

Questions/topics repeatedly answered incorrectly.

Indicators:

- Weak understanding
 - Needs remediation
 - High academic risk
-

FAIR Mastery

Moderate understanding.

Indicators:

- Inconsistent answers
 - Needs improvement
 - Partial competency
-

HIGH Mastery

Questions/topics consistently answered correctly.

Indicators:

- Strong competency
 - Good retention
 - Exam readiness
-

What the System Should Track

Per student:

- Correct answers
- Incorrect answers
- Attempts per question
- Difficulty progression
- Weak topics
- Strong topics
- Repeated failures
- Completion progression
- Improvement trends

Suggested Future Database Models

StudentProgress

Tracks overall student progression.

Suggested Fields

```
id
studentId
currentDifficulty
highestUnlockedDifficulty
completedLevels
overallMastery
lastAttemptDate
```

StudentQuestionPerformance

Tracks performance per question.

Suggested Fields

```
id
studentId
questionId
attempts
```

```
correctAttempts
incorrectAttempts
masteryLevel
lastAnsweredAt
```

ExamAttempt

Tracks individual exam sessions.

Suggested Fields

```
id
studentId
examId
score
percentage
passed
startedAt
submittedAt
```

ExamViolation

Tracks monitoring incidents during exams.

Suggested Fields

```
id
studentId
examId
type
severity
timestamp
resolved
reviewedBy
```

Question Architecture

Questions should eventually support:

Field	Purpose
topic	Academic grouping
category	Subject classification
competency	Learning outcome
difficulty	Question complexity
explanation	Review/remediation
tags	Analytics and filtering

Difficulty Design

The system should support:

Exam-Level Difficulty

Example:

```
EASY EXAM
MEDIUM EXAM
HARD EXAM
EXPERT EXAM
```

This controls:

- progression
- unlocking
- exam categorization

Question-Level Difficulty

Questions may still individually vary.

Example:

```
Easy Question
Medium Question
Hard Question
```

inside the same exam.

This enables:

- adaptive balancing
- competency weighting
- future AI recommendations

Academic Integrity Monitoring

The system also supports proctoring and monitoring.

This becomes part of:

- institutional oversight
- exam supervision
- violation review

Possible Future Violations

Violation	Description
Tab Switching	Student leaves exam tab
Multiple Faces	More than one face detected
No Face Detected	Camera visibility issue
Phone Detection	Mobile device detected
Suspicious Audio	Noise detection
Copy/Paste Attempts	Clipboard monitoring
DevTools Opened	Browser inspection attempt
Excessive Movement	Suspicious body movement

Violation Workflow

Recommended Flow

```
graph TD; A[Student Takes Exam] --> B[AI Monitoring Detects Event]; B --> C[Violation Stored]; C --> D[Admin Monitoring Dashboard]; D --> E[Faculty Review]; E --> F[Academic Decision];
```

Important Academic Principle

The system should NOT automatically punish students.

Recommended flow:

```
graph TD; A[AI detects] --> B[Human reviews]; B --> C[Faculty decides academically];
```

This avoids:

- false accusations
- automated bias
- academic disputes

Roles and Responsibilities

Faculty Responsibilities

Faculty owns:

- exam creation
- scheduling
- question management
- grading
- academic decisions
- progression structure

Faculty is the academic authority.

Admin Responsibilities

Admin owns:

- monitoring
- institutional oversight
- violation review
- analytics
- operational supervision
- system management

Admin is the institutional supervisor.

UI Architecture Philosophy

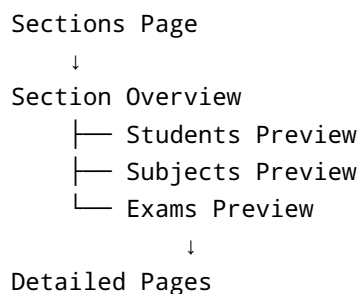
The system follows:

Overview → Detail → Monitoring

instead of:

everything on one page

Current Section Module Architecture



Preview Card Philosophy

Preview cards are:

- operational summaries
- contextual monitoring panels
- navigation gateways

NOT:

- full management systems

Future Exam Monitoring Architecture

Global Exams Page

Institution-wide monitoring.

Shows:

- all exams
- all sections
- all violations
- live activity

Section Exams Page

Contextual monitoring.

Shows:

- section-specific exams
 - section-specific incidents
 - section-specific progression
-

Exam Details Page

Deep operational monitoring.

Future features:

- live student sessions
 - violation timeline
 - risk indicators
 - activity monitoring
 - performance analytics
-

Suggested Future Status System

Exam Status

SCHEDULED
ONGOING
COMPLETED
ARCHIVED

Suggested Future Risk Levels

LOW
MEDIUM
HIGH
CRITICAL

Long-Term Vision

The CNAHS Examination System is evolving into:

Exam Monitoring
+
Adaptive Progression
+
Competency Analytics
+
Academic Integrity Monitoring

This architecture supports:

- competency-based education
- adaptive assessment
- institutional oversight
- AI-assisted monitoring
- progression-driven learning
- board exam preparation

without redesigning the system foundation later.